GaAs Switch-Based Phase Shifter with Wide Bandwidth and Low Loss for 8-Channel Dual-Polarized Reconfigurable Intelligent Surfaces

Heeseung Choi

Electrical and Electronic Engineering, Yonsei University, Seoul, Korea

Introduction

- **Reconfigurable Intelligent Surfaces (RIS)** are a highly promising technology for 6G systems, addressing challenges like signal blockage, coverage extension, and NLOS communication, and requiring a **phase shifter** for effective beam steering.
- This research focuses on implementing this necessary phase shifter for an **8-channel dual-polarized RIS** system using a **GaAs switch** to ensure **low loss and wide bandwidth operation.**

Body

① Reconfigurable Intelligent Surface and Phase Shifter

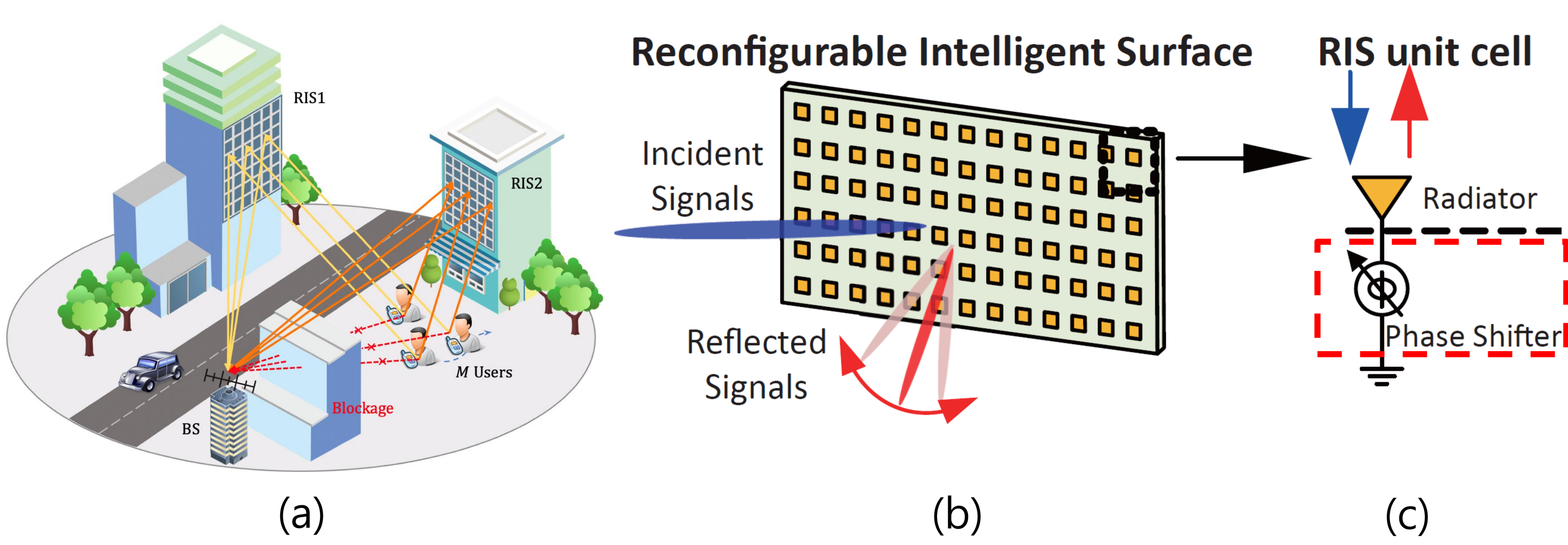


Fig 1. (a) RIS-assisted wireless communication in a blocked environment (b) configuration of RIS and (c) unit cell with phase shifter

② 1-bit Phase Shifter

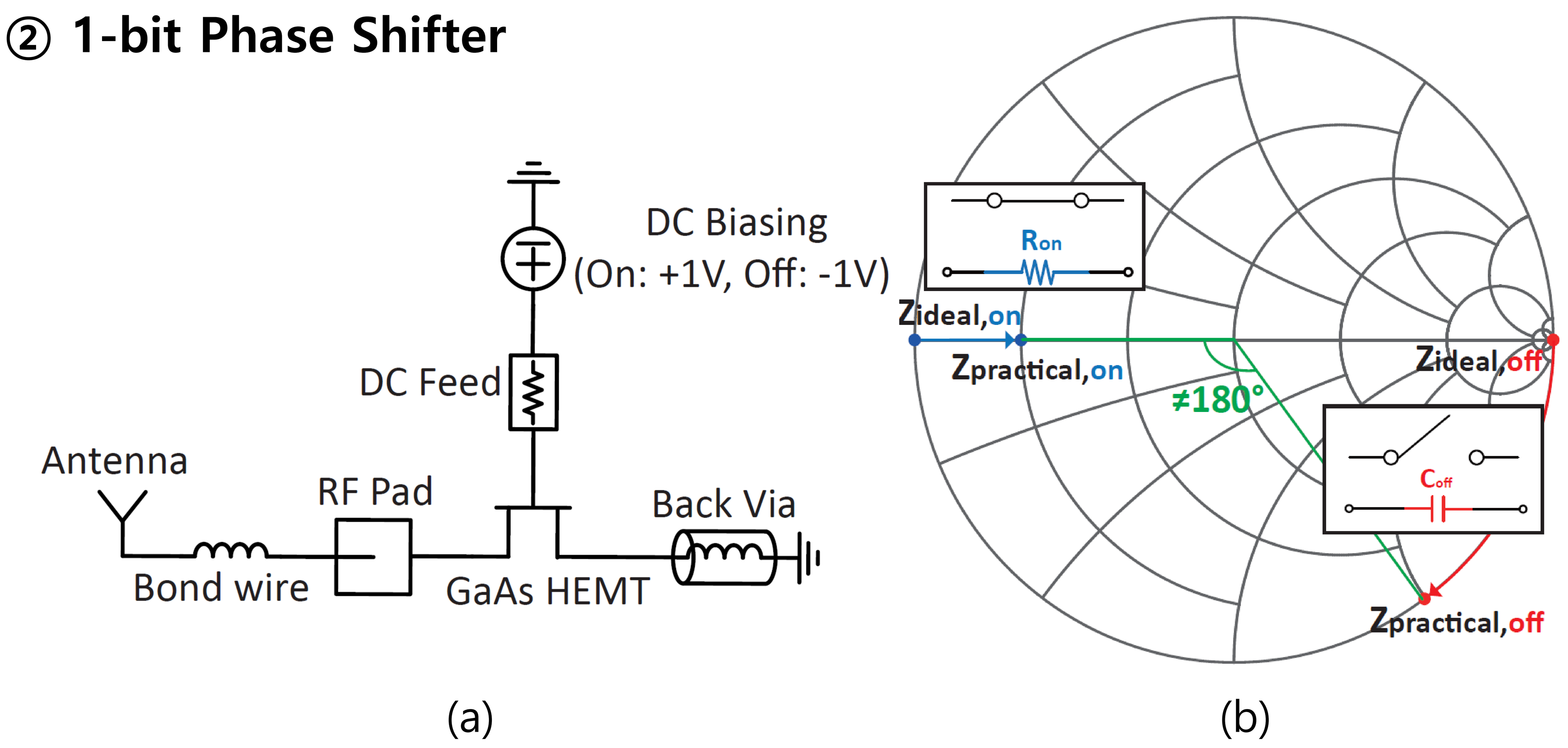
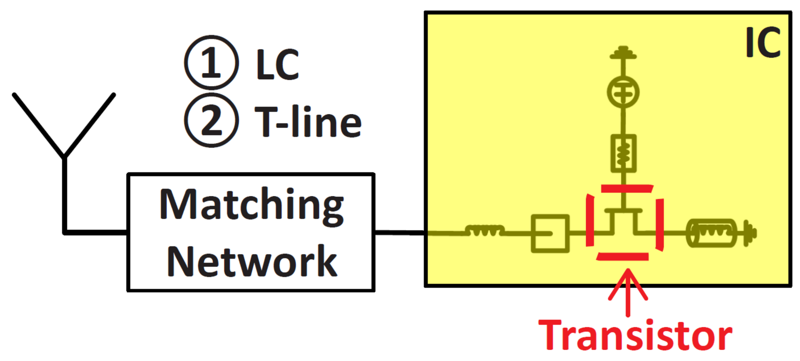


Fig 2. (a) Schematic of the initial 1-bit switch module (b) Ideal and practical impedances on the smith chart



1) Matching Network

→ S_{11} 180° Phase Diff. & Same Mag.

2) Transistor Sizing

→ Min. Loss & Max. Bandwidth

Fig 3. Matching Network design & Transistor Size selection

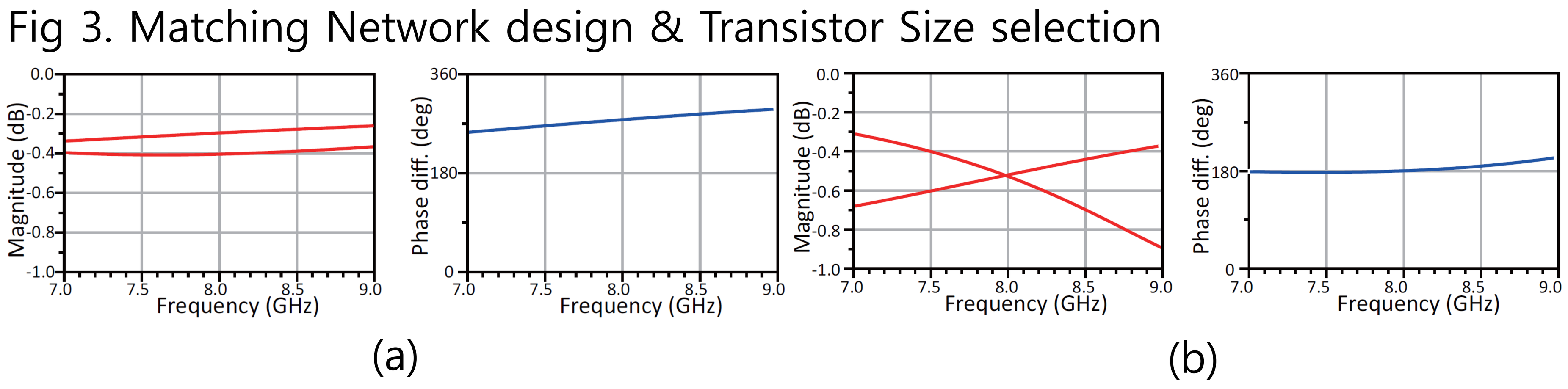


Fig 4. Simulation results of (a) w/o Matching Network (b) w/ Matching Network

③ 1-bit vs. Multi-bit

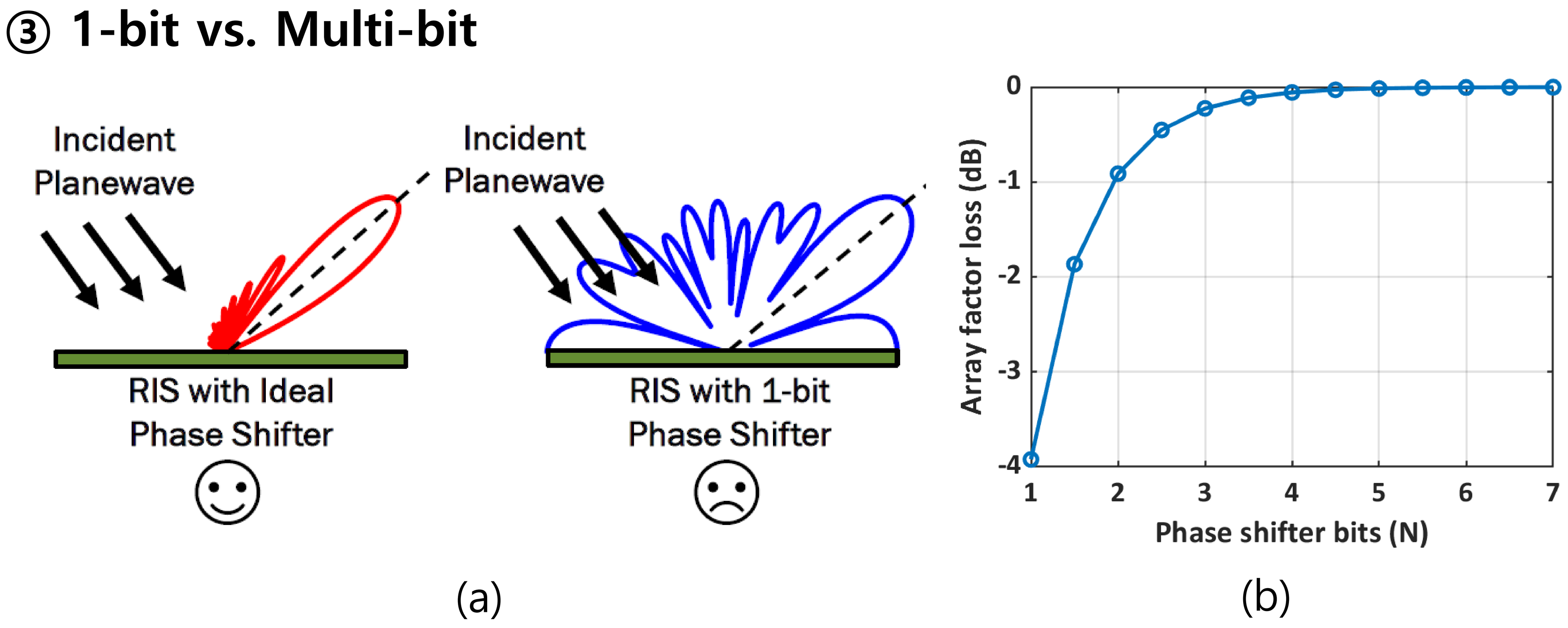
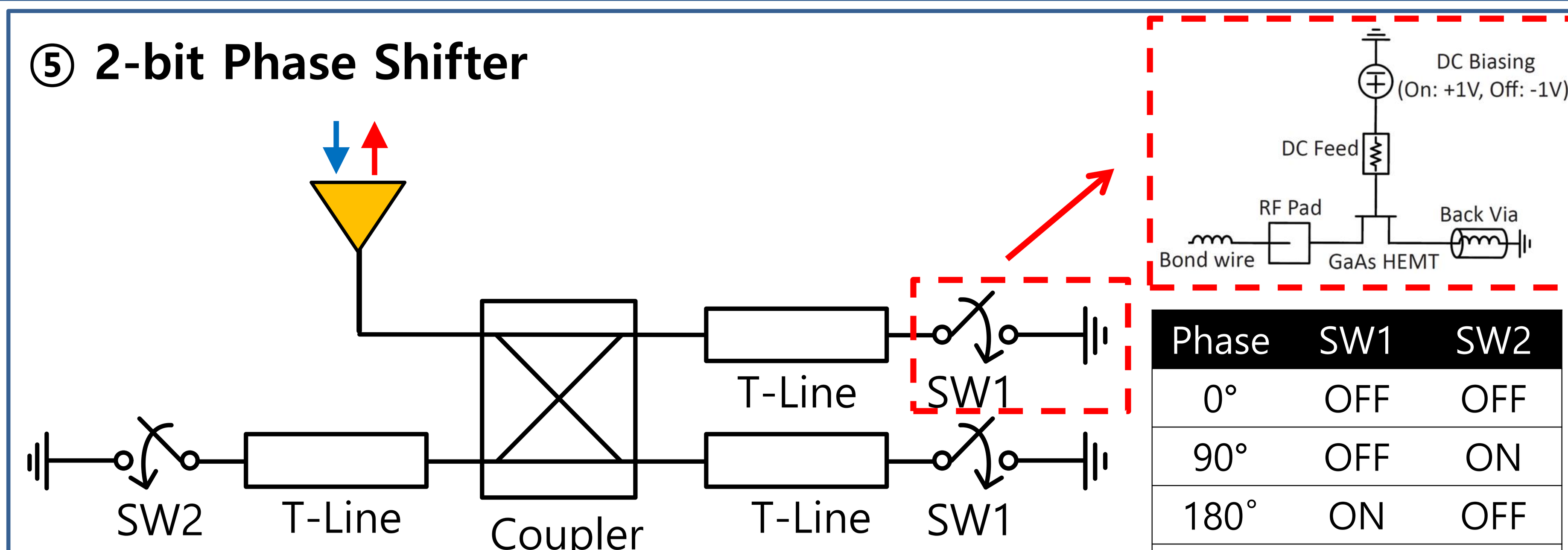


Fig 5. (a) High quantization loss of 1-bit phase shifter (b) Array factor loss vs bits N

⑤ 2-bit Phase Shifter



Phase	SW1	SW2
0°	OFF	OFF
90°	OFF	ON
180°	ON	OFF
270°	ON	ON

Fig 6. Schematic of proposed 2-bit phase shifter

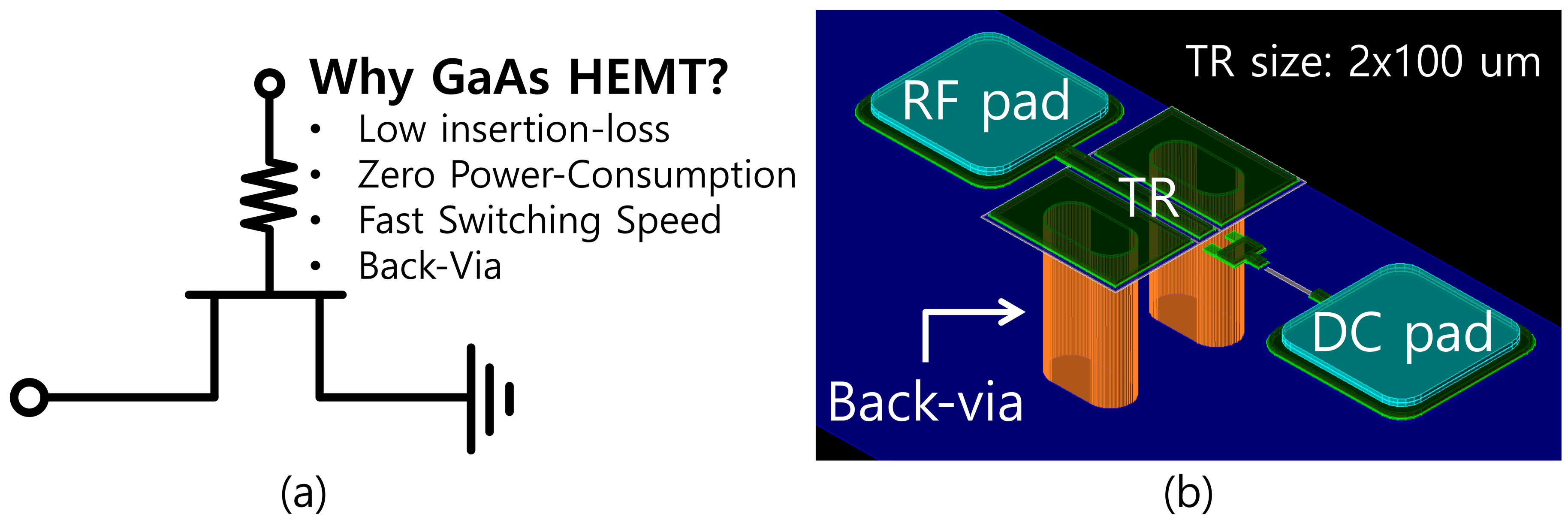


Fig 7. (a) Schematic and (b) layout of a 1-port switch

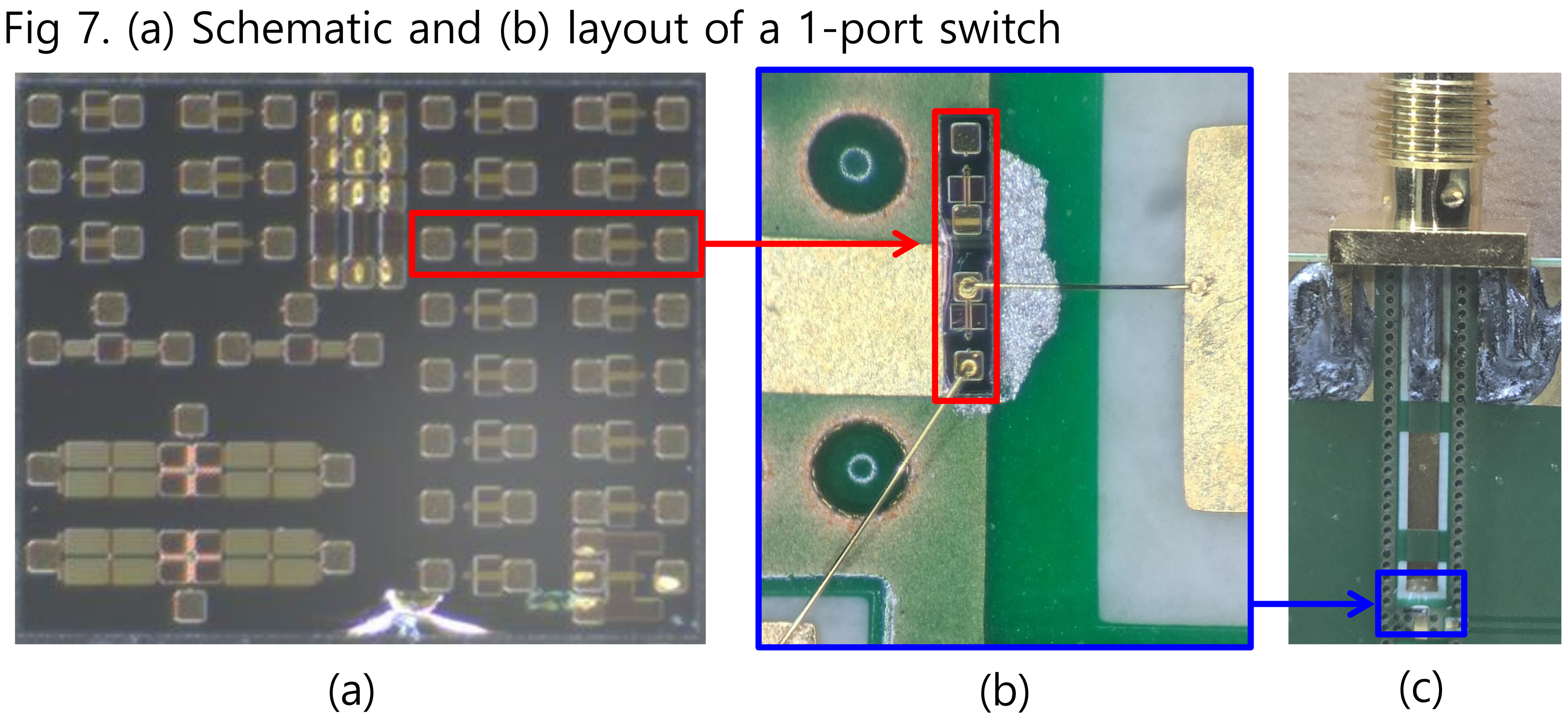


Fig 8. Photograph of (a) die (b) after sawing/wire-bonding (c) chip on board

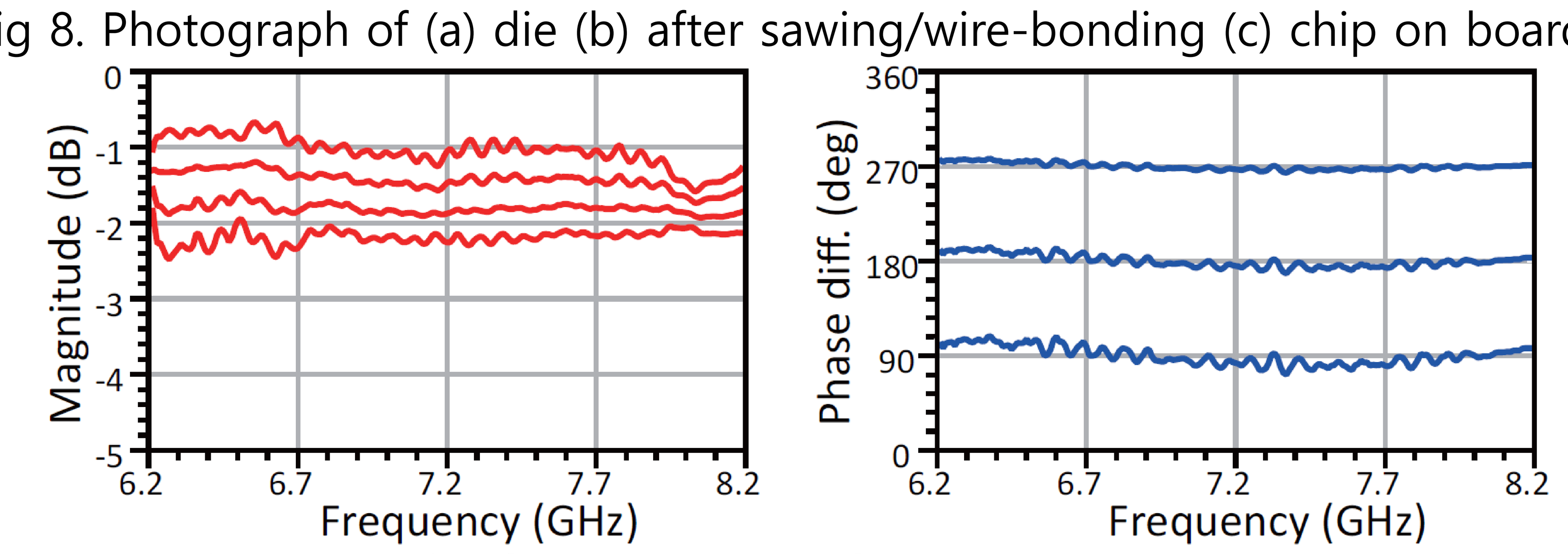


Fig 9. Simulation result of 2-bit phase shifter with measured GaAs switch

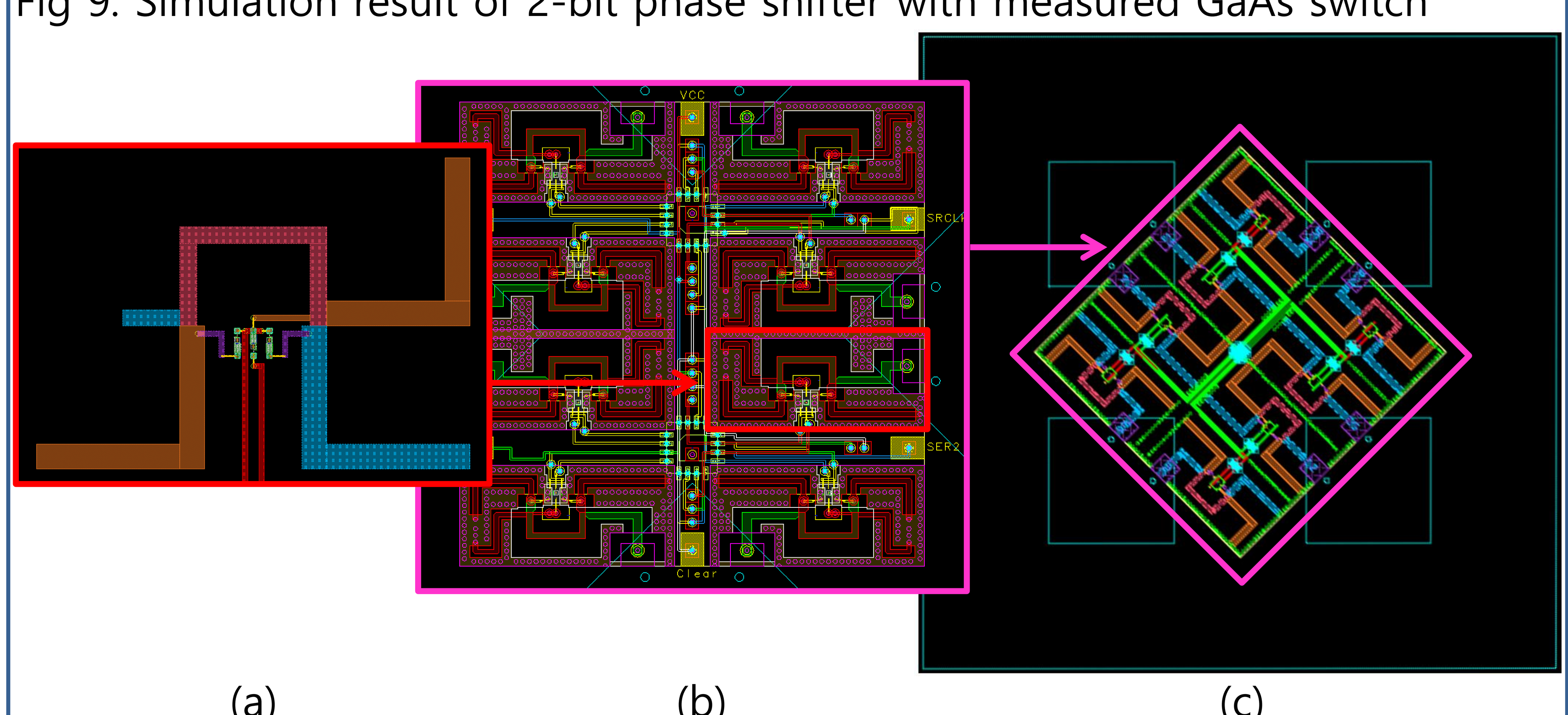


Fig 10. Layout of (a) 2-bit phase shifter in (b) 8-channel module for (c) dual polarized 2x2 unit RIS cell

Conclusion

- **1-bit phase shifter** has a compact structure and simple control, but its high phase-quantization loss limits overall performance.
- **2-bit phase shifter** reduces quantization loss and can be implemented in dual polarized 2x2 unit cell for RIS scaling.
- **Future Work** will focus on 2-bit phase shifter module design and implementation, leveraging the completed fabrication and measurement results of the switch.